

Name: Ayushi Shringi
SID4: 0036
DOMAIN_ID: 4
Domain Entity: Open-source package vulnerabilities
Port: 8036
Prefix: s0036
SEED: 0036
VERIFY_SEED: 260036
Python Version: Python 3.12.14
Embedding Model: sentence-transformers/all-MiniLM-L6-v2
Hardware: MacBook Pro
Chip: Apple M5 Pro
Memory: 24 GB
Operating System: macOS Tahoe 26.5.1
GitHub Repository: <https://github.com/ayushishringi/data260-0036/tree/hw3>
Code Commit Hash: 4194aa905aa3c0116d72beaa382f4d0fac88f4b9
Git Tag: hw3

Part 1 - FastAPI Authentication and Session Management

1. Routes

Route	Method	Purpose
/	GET	Displays the welcome page and navigation links
/login	GET	Displays the login form
/login	POST	Validates credentials and creates a session
/dashboard	GET	Protected page showing the logged-in user
/logout	GET	Clears the session and redirects to the home page

```

def get_current_user(request: Request):
    user = request.session.get("user")
    last_seen = request.session.get("last_seen")

    if user is None or last_seen is None:
        request.session.clear()
        return None

    if time.time() - float(last_seen) > IDLE_TIMEOUT_SECONDS:
        request.session.clear()
        return None

    request.session["last_seen"] = time.time()
    return user

@router.get("/", response_class=HTMLResponse)
async def home(request: Request):
    return templates.TemplateResponse(
        request=request,
        name="home.html",
        context={
            "user": get_current_user(request),
        },
    )

@router.get("/login", response_class=HTMLResponse)
async def login_page(request: Request):
    return templates.TemplateResponse(
        request=request,
        name="login.html",
        context={
            "user": get_current_user(request),
            "error": None,
        },
    )

```

```

3
4 @router.get("/login", response_class=HTMLResponse)
5 async def login_page(request: Request):
6     return templates.TemplateResponse(
7         request=request,
8         name="login.html",
9         context={
10             "user": get_current_user(request),
11             "error": None,
12         },
13     )
14
15
16 @router.post("/login", response_class=HTMLResponse)
17 async def login(
18     request: Request,
19     username: str = Form(...),
20     password: str = Form(...),
21 ):
22     account = USERS.get(username.strip().lower())
23
24     if account is None or account["password"] != password:
25         return templates.TemplateResponse(
26             request=request,
27             name="login.html",
28             context={
29                 "user": None,
30                 "error": "Invalid username or password.",
31             },
32             status_code=401,
33         )
34
35     request.session["user"] = {
36         "username": username.strip().lower(),
37         "name": account["name"],
38     }
39
40     request.session["last_seen"] = time.time()
41
42     return RedirectResponse("/dashboard", status_code=303)
43
44
45 @router.get("/dashboard", response_class=HTMLResponse)
46 async def dashboard(request: Request):
47     # ... (code continues)

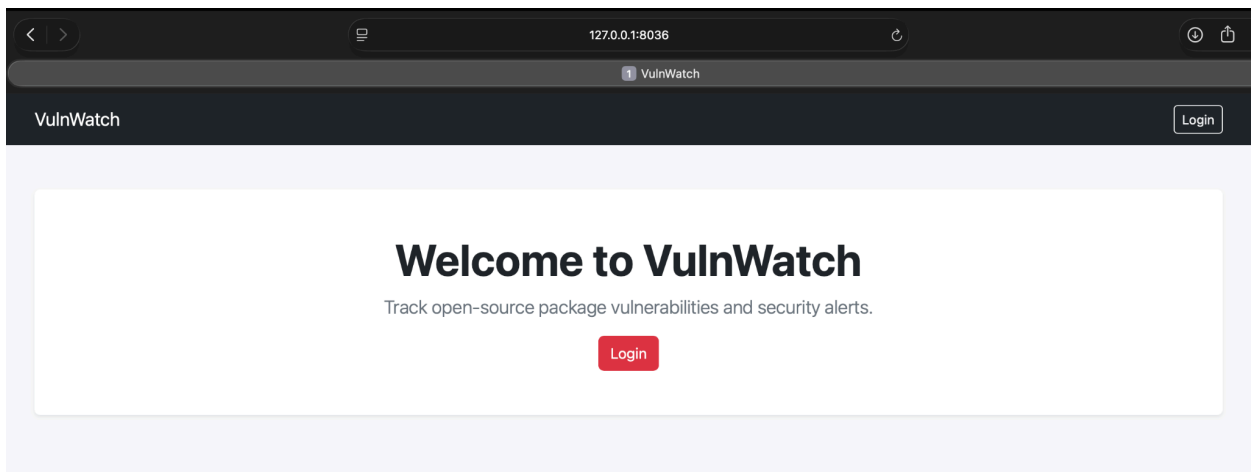
```

```

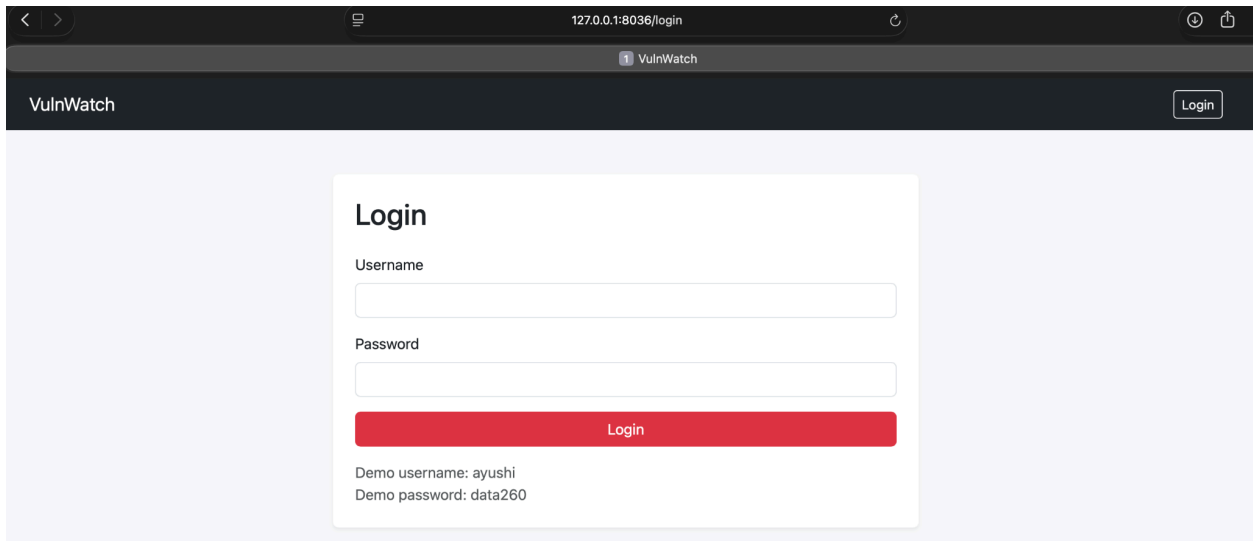
94
95 @router.get("/dashboard", response_class=HTMLResponse)
96 async def dashboard(request: Request):
97     user = get_current_user(request)
98
99     if user is None:
100         return RedirectResponse("/login", status_code=303)
101
102     return templates.TemplateResponse(
103         request=request,
104         name="dashboard.html",
105         context={
106             "user": user,
107         },
108     )
109
110
111 @router.get("/logout")
112 async def logout(request: Request):
113     request.session.clear()
114     return RedirectResponse("/", status_code=303)

```

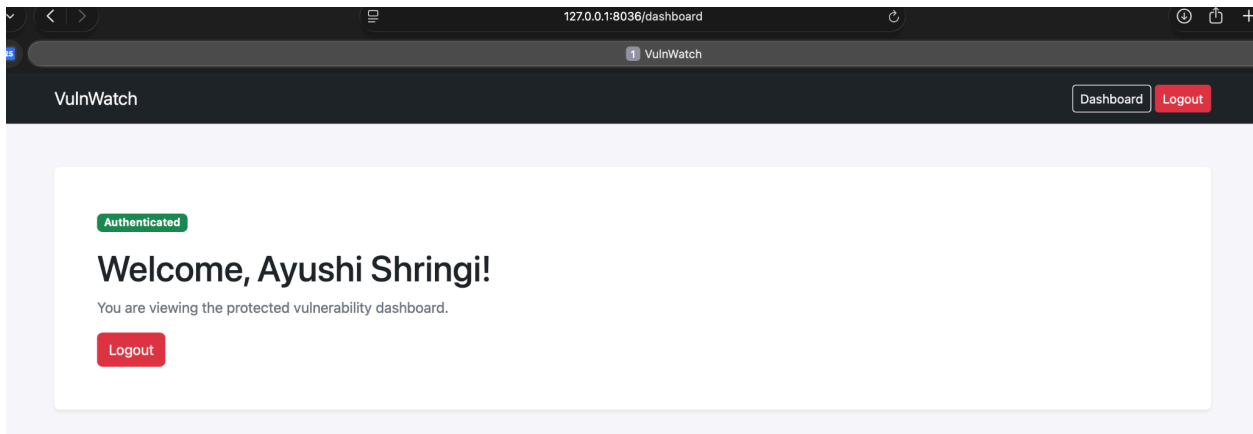
Home Page -



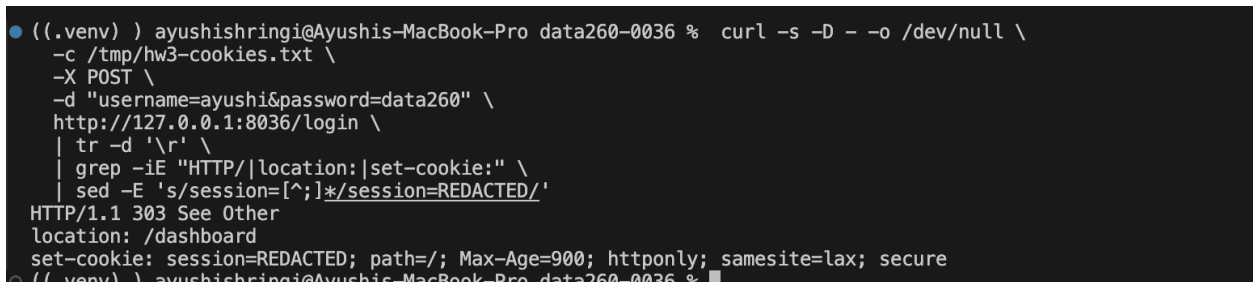
Login Page -



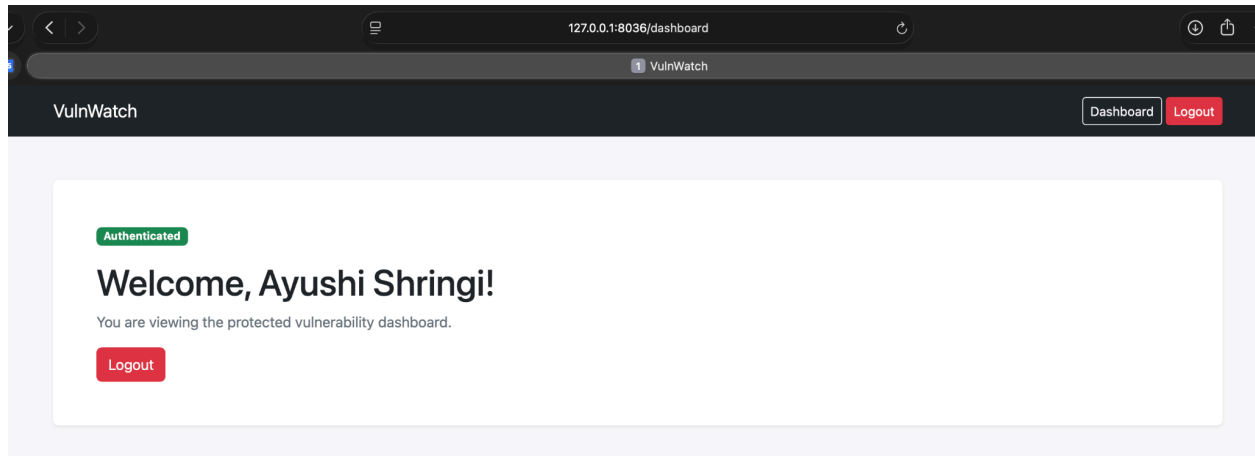
Successful Login -



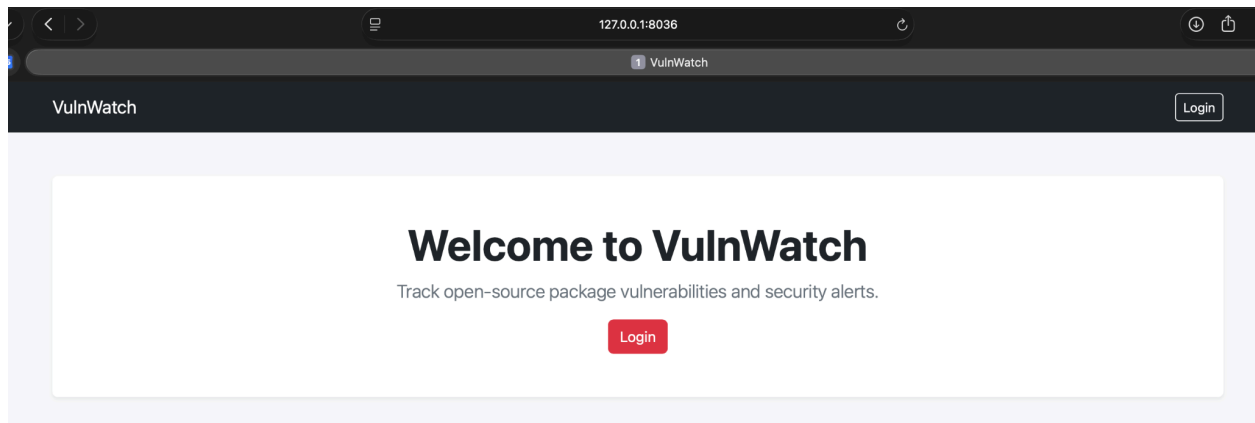
Secure Session Cookie -



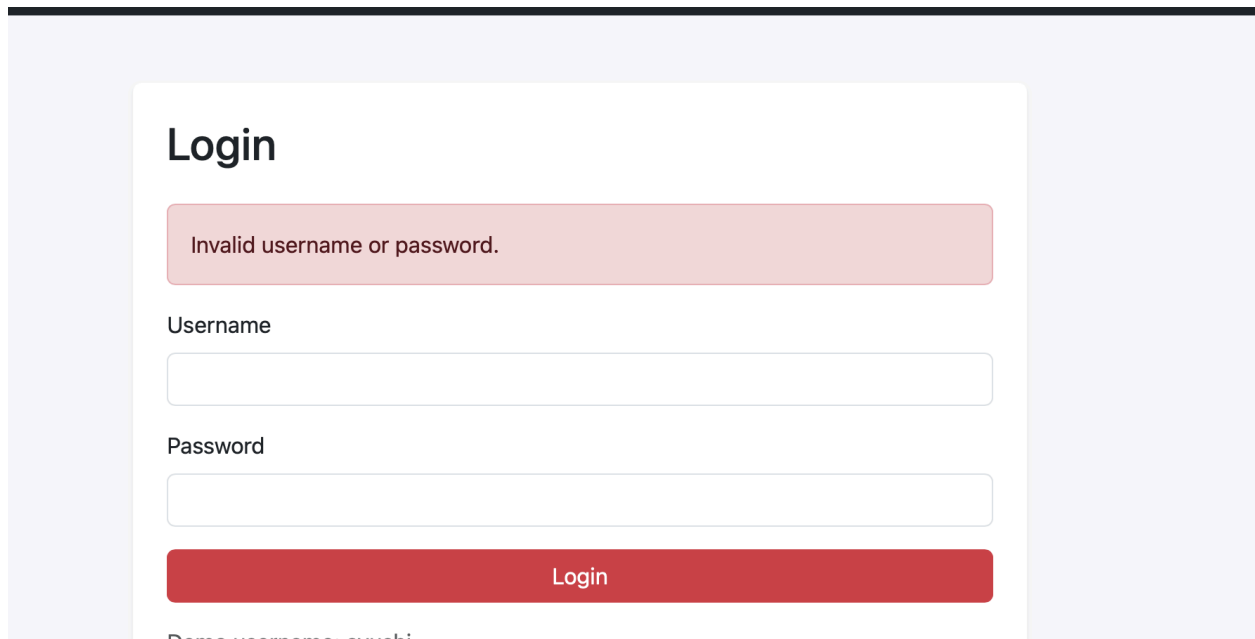
Home Page -



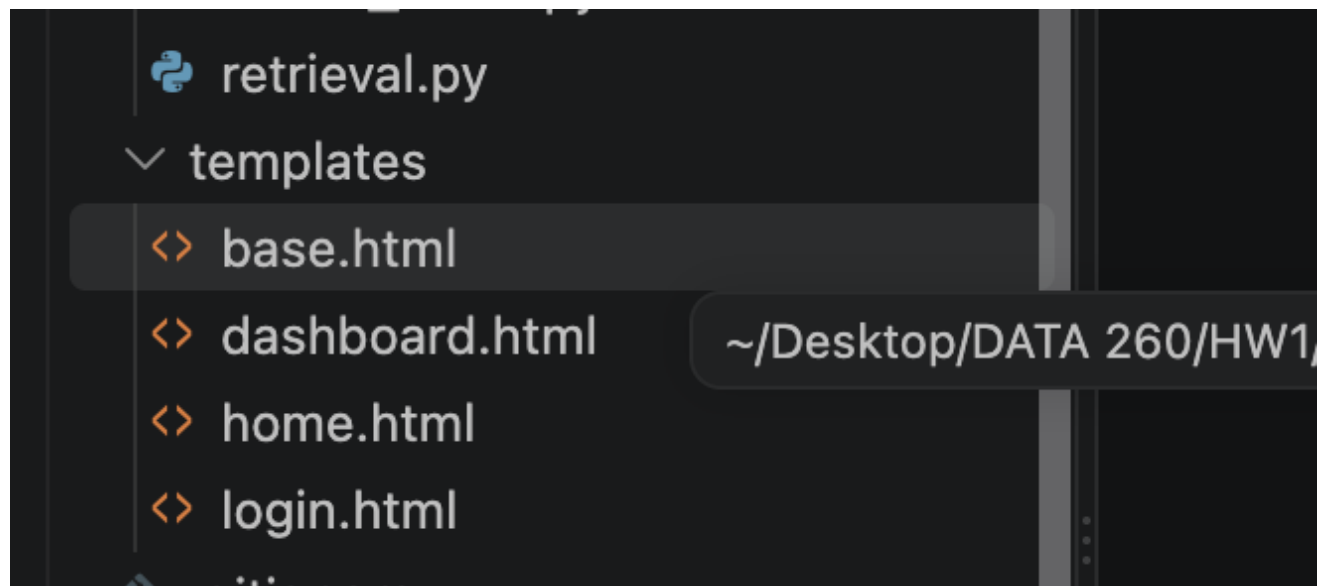
Home Page for Logged out user -



Incorrect Creds -



Templates for authentication page -



```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1">
6
7      <title>{{ title or "VulnWatch" }}</title>
8
9      <link
10         href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
11         rel="stylesheet"
12     >
13 </head>
14
15 <body class="bg-light">
16
17 <nav class="navbar navbar-dark bg-dark">
18     <div class="container">
19         <a class="navbar-brand" href="/">VulnWatch</a>
20
21         <div>
22             {% if user %}
23                 <a class="btn btn-outline-light btn-sm" href="/dashboard">
24                     Dashboard
25                 </a>
26                 <a class="btn btn-danger btn-sm" href="/logout">
27                     Logout
28                 </a>
29             {% else %}
30                 <a class="btn btn-outline-light btn-sm" href="/login">
31                     Login
32                 </a>
33             {% endif %}
34         </div>
35     </div>
36 </nav>
37
38 <main class="container py-5">
39     {% block content %}{% endblock %}
40 </main>
41
42 </body>

```

templates > <> dashboard.html > ...

```
1  {% extends "base.html" %}
2
3  {% block content %}
4
5      <div class="card shadow-sm border-0">
6          <div class="card-body p-5">
7
8              <span class="badge bg-success mb-3">
9                  Authenticated
10             </span>
11
12             <h1>
13                 Welcome, {{ user.name }}!
14             </h1>
15
16             <p class="text-secondary">
17                 You are viewing the protected vulnerability dashboard.
18             </p>
19
20             <a href="/logout" class="btn btn-danger">
21                 Logout
22             </a>
23
24         </div>
25     </div>
26
27 {% endblock %}
```

templates > <> home.html > ...

```
1  {% extends "base.html" %}
2
3  {% block content %}
4
5  <div class="card shadow-sm border-0">
6      <div class="card-body p-5 text-center">
7          <h1 class="display-5 fw-bold">
8              Welcome to VulnWatch
9          </h1>
10
11         <p class="lead text-secondary">
12             Track open-source package vulnerabilities and security alerts.
13         </p>
14
15         {% if user %}
16             <a href="/dashboard" class="btn btn-danger">
17                 Go to Dashboard
18             </a>
19         {% else %}
20             <a href="/login" class="btn btn-danger">
21                 Login
22             </a>
23         {% endif %}
24     </div>
25 </div>
26
27 {% endblock %}
```

```

3  {% block content %}
4
5  <div class="row justify-content-center">
6      <div class="col-md-6">
7
8          <div class="card shadow-sm border-0">
9              <div class="card-body p-4">
10
11                  <h2 class="mb-4">Login</h2>
12
13                  {% if error %}
14                      <div class="alert alert-danger">
15                          {{ error }}
16                      </div>
17                  {% endif %}
18
19                  <form method="post" action="/login">
20
21                      <div class="mb-3">
22                          <label for="username" class="form-label">
23                              Username
24                          </label>
25
26                          <input
27                              id="username"
28                              name="username"
29                              class="form-control"
30                              required
31                          >
32                      </div>
33
34                      <div class="mb-3">
35                          <label for="password" class="form-label">
36                              Password
37                          </label>
38
39                          <input
40                              id="password"
41                              name="password"
42                              type="password"
43                              class="form-control"
44                              required
45                          >
46                      </div>
47
48                      <button type="submit" class="btn btn-danger w-100">
49                          Login
50                      </button>
51                  </form>
52

```

Expired session redirected to login -

```
((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % curl -i -b /tmp/hw3-timeout.txt \
http://127.0.0.1:8036/dashboard
HTTP/1.1 303 See Other
date: Mon, 21 Sep 2026 18:25:59 GMT
server: uvicorn
content-length: 0
location: /login
vary: Cookie
set-cookie: session=null; path=/; expires=Thu, 01 Jan 1970 00:00:00 GMT; httponly; samesite=lax

((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 %
```

Three LlamaIndex chunking techniques

```

c > retrieval.py > query_indexes
9 |     TokenTextSplitter,
10 | )
11 | from llama_index.embeddings.huggingface import HuggingFaceEmbedding
12 |
13 |
14 | ROOT = Path(__file__).resolve().parents[1]
15 | CORPUS_FILE = ROOT / "corpus" / "github_reviewed_advisories_snapshot.txt"
16 |
17 | MODEL_NAME = "sentence-transformers/all-MiniLM-L6-v2"
18 |
19 |
20 | def load_corpus() -> list[Document]:
21 |     text = CORPUS_FILE.read_text(encoding="utf-8")
22 |
23 |     if len(text.encode("utf-8")) < 200_000:
24 |         raise ValueError("Corpus must be at least 200 KB.")
25 |
26 |     return [
27 |         Document(
28 |             text=text,
29 |             metadata={
30 |                 "source": "GitHub Advisory Database",
31 |                 "corpus_file": str(CORPUS_FILE),
32 |             },
33 |         )
34 |     ]
35 |
36 |
37 | def average_chunk_length(nodes) -> float:
38 |     if not nodes:
39 |         return 0.0
40 |
41 |     lengths = [
42 |         len(node.get_content())
43 |         for node in nodes
44 |     ]
45 |
46 |     return round(float(np.mean(lengths)), 2)
47 |
48 |
49 | def create_pipeline(nodes, embed_model):
50 |     return {
51 |         "index": VectorStoreIndex(
52 |             nodes,
53 |             embed_model=embed_model,
54 |         ),

```

```

49 def create_pipeline(nodes, embed_model):
50     nodes : nodes,
51     "embed_model": embed_model,
52     "chunk_count": len(nodes),
53     "avg_chunk_length": average_chunk_length(nodes),
54 }
55
56
57
58
59
60
61
62 def build_indexes():
63     embed_model = HuggingFaceEmbedding(
64         model_name=MODEL_NAME
65     )
66
67     Settings.embed_model = embed_model
68
69     documents = load_corpus()
70
71     token_parser = TokenTextSplitter(
72         chunk_size=512,
73         chunk_overlap=50,
74     )
75
76     token_nodes = token_parser.get_nodes_from_documents(documents)
77
78     semantic_parser = SemanticSplitterNodeParser(
79         buffer_size=1,
80         breakpoint_percentile_threshold=95,
81         embed_model=embed_model,
82     )
83
84     semantic_nodes = semantic_parser.get_nodes_from_documents(documents)
85
86     window_parser = SentenceWindowNodeParser.from_defaults(
87         window_size=3,
88         window_metadata_key="window",
89         original_text_metadata_key="original_text",
90         include_prev_next_rel=True,
91     )
92
93     window_nodes = window_parser.get_nodes_from_documents(documents)
94
95     return {
96         "token": create_pipeline(
97             token_nodes,
98             embed_model,
99         ),

```

```

103         ),
104         "sentence_window": create_pipeline(
105             window_nodes,
106             embed_model,
107         ),
108     }
109
110
111     def cosine_similarity(query_vector, document_vector) -> float:
112         query_norm = np.linalg.norm(query_vector)
113         document_norm = np.linalg.norm(document_vector)
114
115         if query_norm == 0 or document_norm == 0:
116             return 0.0
117
118         return float(
119             np.dot(query_vector, document_vector)
120             / (query_norm * document_norm)
121         )
122
123
124     def query_indexes(indexes, question: str, top_k: int = 3) -> dict:
125         results = {}
126
127         for name, pipeline in indexes.items():
128             embed_model = pipeline["embed_model"]
129
130             query_embedding = np.asarray(
131                 embed_model.get_query_embedding(question),
132                 dtype=np.float32,
133             )
134
135             retriever = pipeline["index"].as_retriever(
136                 similarity_top_k=top_k
137             )
138
139             started = time.perf_counter()
140             source_nodes = retriever.retrieve(question)
141             retrieval_latency_ms = (
142                 time.perf_counter() - started
143             ) * 1000
144
145             retrieved_rows = []
146             document_vectors = []
147

```

```

124 def query_indexes(indexes, question: str, top_k: int = 3) -> dict:
153
154     document_embedding = np.asarray(
155         embed_model.get_text_embedding(text),
156         dtype=np.float32,
157     )
158
159     document_vectors.append(document_embedding)
160
161     retrieved_rows.append(
162         {
163             "rank": rank,
164             "store_score": (
165                 float(node_with_score.score)
166                 if node_with_score.score is not None
167                 else None
168             ),
169             "cosine_similarity": round(
170                 cosine_similarity(
171                     query_embedding,
172                     document_embedding,
173                 ),
174                 6,
175             ),
176             "chunk_length": len(text),
177             "preview": text[:160].replace("\n", " "),
178         }
179     )
180
181     if document_vectors:
182         document_matrix = np.vstack(document_vectors)
183         document_shape = list(document_matrix.shape)
184     else:
185         document_shape = [0, int(query_embedding.shape[0])]
186
187     results[name] = {
188         "technique": name,
189         "question": question,
190         "chunk_count": pipeline["chunk_count"],
191         "avg_chunk_length": pipeline["avg_chunk_length"],
192         "query_embedding_dimension": int(
193             query_embedding.shape[0]
194         ),
195         "query_embedding_first_8": [
196             round(float(value), 8)
197             for value in query_embedding[:8]

```

Detailed Retrieval Output-

```
1  [
2  {
3    "question_id": "q1",
4    "question": "Which packages are affected by prototype pollution vulnerabilities?",
5    "expected_answer": "The web3-utils package is vulnerable before version 4.2.1 because of insecure recursive merge behavior.",
6    "expected_source_file": "corpus/github_reviewed_advisories_snapshot.txt",
7    "expected_source_match": "web3-utils",
8    "results": {
9      "token": {
10       "technique": "token",
11       "question": "Which packages are affected by prototype pollution vulnerabilities?",
12       "chunk_count": 1737,
13       "avg_chunk_length": 1091.45,
14       "query_embedding_dimension": 384,
15       "query_embedding_first_8": [
16         -0.06461962,
17         0.04836296,
18         0.09498411,
19         -0.06370848,
20         0.09666511,
21         -0.00856569,
22         0.01431219,
23         0.03795227
24       ],
25       "query_vector_shape": [
26         384
27       ],
28       "document_vectors_shape": [
29         3,
30         384
31       ],
32       "retrieval_latency_ms": 18.8241,
33       "source_nodes": [
34         {
35           "rank": 1,
36           "store_score": 0.5635228042109589,
37           "cosine_similarity": 0.502413,
38           "chunk_length": 1096,
39           "preview": "This link is maintained to preserve external references.\\n\\n## Original Description\\nVersions of the package web3-utils before 4.
40         },
41         {
42           "rank": 2,
43           "store_score": 0.4015562437466903,
44           "cosine_similarity": 0.394947,
45           "chunk_length": 1039,
46           "preview": "and `0.10.4` we've been unable to easily assess if these heavily outdated versions are vulnerable or not to this exploit.\\n\\n## S
47         },
48         {
49           "rank": 3,
50           "store_score": 0.39343514328723544,
51           "cosine_similarity": 0.383106,
52           "chunk_length": 1511,
53           "preview": "Mitigation factors\\n\\nMicrosoft has not identified any mitigating factors for this vulnerability.\\n\\n## Affected softwar
54         }
55       ]
56     }
57   }
58 ]
```

```

    }
  }
},
"semantic": {
  "technique": "semantic",
  "question": "Which packages are affected by prototype pollution vulnerabilities?",
  "chunk_count": 93,
  "avg_chunk_length": 18464.8,
  "query_embedding_dimension": 384,
  "query_embedding_first_8": [
    -0.06461962,
    0.04836296,
    0.09498411,
    -0.06370848,
    0.09666511,
    -0.00856569,
    0.01431219,
    0.03795227
  ],
  "query_vector_shape": [
    384
  ],
  "document_vectors_shape": [
    3,
    384
  ],
  "retrieval_latency_ms": 3.6955,
  "source_nodes": [
    {
      "rank": 1,
      "store_score": 0.3664111535218756,
      "cosine_similarity": 0.35252,
      "chunk_length": 7143,
      "preview": "This vulnerability also applies to the Pulsar Broker when it is configured with \\\"\\\"functionsWorkerEnabled=true\\\"\\\".\\n\\nThis issue a",
    },
    {
      "rank": 2,
      "store_score": 0.3607522753550862,
      "cosine_similarity": 0.294423,
      "chunk_length": 4559,
      "preview": "Being able to call some of these methods can have security implications.\\n\\n### Details\\nCommands verify that the class must be a",
    },
    {
      "rank": 3,
      "store_score": 0.3391151835674793,
      "cosine_similarity": 0.296239,
      "chunk_length": 10478,
      "preview": "For some applications, invalid tokens may be ignored or cause a warning or a crash. On the other hand, for applications that store pos",
    }
  ]
}
},

```

```

105 "sentence_window": {
106   "technique": "sentence_window",
107   "question": "Which packages are affected by prototype pollution vulnerabilities?",
108   "chunk_count": 1834,
109   "avg_chunk_length": 936.33,
110   "query_embedding_dimension": 384,
111   "query_embedding_first_8": [
112     -0.06461962,
113     0.04836296,
114     0.09498411,
115     -0.06370848,
116     0.09666511,
117     -0.00856569,
118     0.01431219,
119     0.03795227
120   ],
121   "query_vector_shape": [
122     384
123   ],
124   "document_vectors_shape": [
125     3,
126     384
127   ],
128   "retrieval_latency_ms": 15.2547,
129   "source_nodes": [
130     {
131       "rank": 1,
132       "store_score": 0.6269334882292794,
133       "cosine_similarity": 0.630736,
134       "chunk_length": 465,
135       "preview": "This link is maintained to preserve external references.\\n\\n# Original Description\\n\\nVersions of the package web3-utils before 4.2.
136     },
137     {
138       "rank": 2,
139       "store_score": 0.42291045926147436,
140       "cosine_similarity": 0.405839,
141       "chunk_length": 215,
142       "preview": "***SDK Users:** Developers integrating Nuclei into their platforms, particularly if they permit the execution of custom workflows by er
143     },
144     {
145       "rank": 3,
146       "store_score": 0.40579703272261236,
147       "cosine_similarity": 0.39694,
148       "chunk_length": 2280,
149       "preview": "An attacker has the ability to modify an object's prototype, which could result in changing the behavior of all objects that inherit f
150     }
151   ]
152 }
153 }
154 },

```

Retrieval outputs and comparison metrics -

```
((.venv) ) ayushishringi@Ayushis-MacBook-Pro data260-0036 % .venv/bin/python scripts/run_hw3_retrieval.py  
Loading weights: 100%|██████████████████████████████████████████████████████████████| 103/103 [00:00<00:00, 9824.51it/  
  
Question q1: Which packages are affected by prototype pollution vulnerabilities?  
token: 3 chunks, latency=18.9942 ms  
semantic: 3 chunks, latency=5.7132 ms  
sentence_window: 3 chunks, latency=14.8791 ms  
  
Question q2: What versions of Apache Pulsar are affected by security vulnerabilities?  
token: 3 chunks, latency=14.3226 ms  
semantic: 3 chunks, latency=3.9078 ms  
sentence_window: 3 chunks, latency=14.9849 ms  
  
Question q3: What are common mitigation steps described in the advisories?  
token: 3 chunks, latency=14.0733 ms  
semantic: 3 chunks, latency=4.0662 ms  
sentence_window: 3 chunks, latency=14.4565 ms  
  
Question q4: Which vulnerabilities allow remote code execution?  
token: 3 chunks, latency=14.4575 ms  
semantic: 3 chunks, latency=4.6062 ms  
sentence_window: 3 chunks, latency=14.2827 ms  
  
Question q5: Which advisories mention denial of service?  
token: 3 chunks, latency=14.5475 ms  
semantic: 3 chunks, latency=4.2342 ms  
sentence_window: 3 chunks, latency=14.8704 ms  
  
Summary metrics:  
{  
  "token": {  
    "questions_evaluated": 5,  
    "chunks_produced": 1737,  
    "average_chunk_length": 1091.45,  
    "top_1_cosine_mean": 0.484747,  
    "mean_at_k_cosine": 0.427933,  
    "recall_at_k": 0.6,  
    "mean_retrieval_latency_ms": 15.27902  
  },  
  "semantic": {  
    "questions_evaluated": 5,  
    "chunks_produced": 93,  
    "average_chunk_length": 18464.8,  
    "top_1_cosine_mean": 0.405813,  
    "mean_at_k_cosine": 0.333585,  
    "recall_at_k": 0.2,  
    "mean_retrieval_latency_ms": 4.50552  
  },  
  "sentence_window": {  
    "questions_evaluated": 5,  
    "chunks_produced": 1834,  
    "average_chunk_length": 936.33,  
    "top_1_cosine_mean": 0.534835,  
    "mean_at_k_cosine": 0.479562,  
    "recall_at_k": 0.4,  
    "mean_retrieval_latency_ms": 14.69472  
  }  
}
```

```

    "mean_retrieval_latency_ms": 4.02554
  },
  "sentence_window": {
    "questions_evaluated": 5,
    "chunks_produced": 1834,
    "average_chunk_length": 936.33,
    "top_1_cosine_mean": 0.534835,
    "mean_at_k_cosine": 0.479562,
    "recall_at_k": 0.4,
    "mean_retrieval_latency_ms": 14.67206
  }
}

```

Retrieval-quality comparison -

```

1  # HW3 Retrieval Metrics
2
3  ## Configuration
4
5  - Domain ID: 4
6  - Domain: Open-source package vulnerabilities
7  - Corpus: `corpus/github_reviewed_advisories_snapshot.txt`
8  - Embedding model: `sentence-transformers/all-MiniLM-L6-v2`
9  - Questions evaluated: 5
10 - Retrieval depth: k = 3
11 - Retrieval mode: Retrieval-only; no external LLM was used
12
13 ## Retrieval Quality Comparison
14
15 | Technique | Chunks | Average chunk length | Top-1 cosine | Mean@3 cosine | Recall@3 | Mean retrieval latency (ms) |
16 |---|---|---|---|---|---|---|
17 | Token | 1737 | 1091.45 | 0.484747 | 0.427933 | 0.60 | 16.586620 |
18 | Semantic | 93 | 18464.80 | 0.405813 | 0.333585 | 0.20 | 4.152180 |
19 | Sentence-window | 1834 | 936.33 | 0.534835 | 0.479562 | 0.40 | 15.155020 |
20
21 ## Interpretation
22
23 Sentence-window chunking produced the strongest cosine-similarity results. It achieved the highest average top-1 cosine score and the highest mean@3 cosine score. Its sentence-level chunks preserved enough local context while remaining focused on individual advisory statements.
24
25 Token chunking achieved the highest Recall@3 using the expected source-match check. It produced moderately sized chunks and performed consistently across the five questions.
26
27 Semantic chunking was the fastest at retrieval time and produced far fewer chunks. However, its average chunk length was much larger, and its cosine and recall values were lower. Large semantic chunks may contain several unrelated advisory sections, which can reduce retrieval precision for focused questions.
28
29 ## Conclusion
30
31 For this vulnerability-advisory corpus, Sentence-window chunking is the best overall technique based on cosine similarity. Token chunking is a strong alternative when source-match recall is prioritized. Semantic chunking is fastest but less precise for these questions because its chunks are very large.
32
33 ## Reproducibility
34
35 The raw per-query and per-technique results are stored in:
36
37 ```text
38 reports/hw03/raw/retrieval_outputs.json

```

Retrieval-Only RAG with Three LlamaIndex Chunking Techniques -

The second part compares three LlamaIndex chunking techniques for retrieval-only RAG over a local corpus of open-source package vulnerability advisories. The corpus

contains 500 GitHub-reviewed advisory records and is 1,717,675 bytes, which exceeds the required 200 KB minimum.

The embedding model used for all three techniques was ``sentence-transformers/all-MiniLM-L6-v2``. Each technique produced an in-memory `VectorStoreIndex`. The retrieval function computed the query embedding, retrieved the top three nodes, calculated cosine similarity with each returned document embedding, recorded chunk lengths, and measured retrieval latency.

1. Chunking Techniques - Token-based chunking used ``TokenTextSplitter`` with a chunk size of 512 tokens and an overlap of 50 tokens. Semantic chunking used ``SemanticSplitterNodeParser`` with a buffer size of 1 and a breakpoint percentile threshold of 95. Sentence-window chunking used ``SentenceWindowNodeParser`` with a three-sentence window and neighboring sentence relationships enabled.

2. Retrieval Results - Sentence-window chunking produced the strongest cosine-similarity results, with a mean top-1 cosine score of 0.534835 and a mean@3 cosine score of 0.479562. Token chunking achieved the highest Recall@3 value of 0.60. Semantic chunking was the fastest technique, with a mean retrieval latency of 4.15218 milliseconds, but it produced very large chunks and had the lowest cosine and recall values.

3. Observations - Sentence-window chunking performed well because it preserved sentence-level meaning while retaining nearby context. This was useful for vulnerability advisories, where the affected package, vulnerability behavior, and mitigation information are often described across neighboring sentences.

Token chunking provided a useful balance between chunk size and retrieval coverage. Semantic chunking created only 93 chunks, but the average chunk length was 18,464.80 characters. These very large chunks sometimes combined unrelated advisory sections, which likely reduced retrieval precision.

4. Incorrect High-Scored Retrieval - At least one high-scored retrieval did not contain the expected answer. For example, the prototype-pollution question returned unrelated vulnerability text for some techniques, including sections about Apache Pulsar and other packages instead of the expected web3-utils advisory. This likely occurred because the embedding model recognized general security and vulnerability language as semantically similar even though the retrieved chunk did not contain the specific package requested.

5. Conclusion - For this open-source package vulnerability corpus, Sentence-window chunking is the best overall technique based on the highest top-1 and mean@3 cosine scores. Token chunking is a strong alternative when recall is the main priority. Semantic chunking is fastest, but its very large chunks reduce precision for focused vulnerability questions.

Reproducible Run Instructions -

From the repository root, activate the virtual environment:

```
source .venv/bin/activate
```

Start the FastAPI application:

```
SESSION_HTTPS_ONLY=false .venv/bin/python -m uvicorn backend.main:app --reload  
--port 8036
```

In a second terminal, run:

```
.venv/bin/python scripts/build_corpus_manifest.py
```

```
.venv/bin/python scripts/run_hw3_retrieval.py
```

```
.venv/bin/python scripts/verify_hw03.py
```

The first command starts the FastAPI application. The remaining commands rebuild the corpus manifest, run the three retrieval pipelines, save the raw retrieval results and metrics, and verify the HW3 implementation.

AI Use Disclosure

I used an AI assistant to understand the HW3 requirements, plan the authentication and retrieval implementation, debug errors, and review the retrieval metrics. I personally edited the files in VS Code, ran the commands, tested the application, verified the outputs, and committed the work.

An initial retrieval implementation incorrectly used LlamaIndex's default OpenAI query engine and failed because no `OPENAI\_API\_KEY` was configured. I identified this from the traceback and changed the implementation to retrieval-only mode using `as\_retriever()`. This removed the external LLM dependency and allowed the application to run locally.

Retrieval-Quality Comparison -

Technique	Chunks	Avg chunk length	Top-1 cosine	Mean@3 cosine	Recall @3	Mean latency
Token	1737	1091.45	0.484747	0.427933	0.60	16.586620
Semantic	93	18464.80	0.405813	0.333585	0.20	4.152180
Sentence-window	1834	936.33	0.534835	0.479562	0.40	15.155020